David Antwi

david.antwi@yale.edu | Personal Website | LinkedIn | github.com/Alienware2000

Education

Yale University New Haven, CT
B.S. Electrical Engineering and Computer Science *Expected May 2028*
- **Coursework:** Data Structures, Systems Programming, Software Engineering, Full-Stack Web, Intro to Machine Learning, Intro to AI, HCI, Linear Algebra, Discrete Math

Experience

BENMP Jul 2026 – Present
Software Engineer, Tech Team (AI and Full-Stack) *International organization, 30+ countries*
- Built and launched the organization's partner management platform, now holding 16,000+ partners across 30+ countries and used by admins at 117 regional hubs.
- Built mobile money payment reconciliation, plus guardrailed, provider-agnostic WhatsApp automation.
- Built a grounded, read-only AI assistant over partner data: code computes every figure, the model only explains it.
- Archived 26,000+ legacy records with count verification and zero loss, then expanded the hub platform from Ghana to 30+ countries in under a month.

Beakr, Inc. (Stealth) May 2026 – Aug 2026
Software Engineering Intern, AI memory platform for life sciences research *New York, NY*
- Designed and shipped Beakr Interviews, an AI system that finds gaps in a lab's knowledge base, interviews the right researcher by text or voice, and turns their answers into cited wiki entries.
- Led the computer-use desktop agent: rebuilt search over 80K+ files, made every answer cite its source file, and got it running reliably on Windows by fixing 35 release-blocking defects (Rust).
- Extended the desktop agent to launch and supervise coding agents (Claude Code, Codex) on the user's computer, with explicit approval before each run and a full audit log.
- Built the agent's write access to 19 external services, with scoped permissions and user confirmation before every action.

SynovAI Oct 2025 – May 2026
Software Engineering Intern, AI retrosynthesis startup *New Haven, CT*
- Designed and built a vendor-aware stock filtering feature that shows researchers only routes whose starting materials are purchasable, across 5 vendor catalogs of 2.5M+ compounds.
- Built a chemical-safety layer that pulls GHS hazard data from PubChem and flags hazards on each molecule in a route.
- Built a self-serve fine-tuning pipeline (PyTorch, QLoRA) covering validation, training, and automated evaluation.
- Led a UI/UX overhaul of the researcher platform (React, Next.js), redesigning the settings and results views.

Yale Model African Union Nov 2025 – Present
Lead Developer *New Haven, CT*
- Replaced the Squarespace site with a custom Next.js rebuild of yalemodelau.org, plus a registration platform with Stripe payments now used by 4,600+ delegate applicants.
- Built the board's review desk for delegate applications and financial aid, with blind review and role-based access.
- Built the previous conference's delegate training platform, serving 335+ users.

Projects

Intentionality | *AI-Powered Gamified Productivity Platform* 2026
- Launched a gamified productivity app with streaks and friend accountability, used by 300+ people.
- Built an AI planner grounded in users' calendars and coursework, with failover across two LLMs.

Better Office Hours | *1st Place, YaleAI x SpaceXAI Cursor Build Challenge* 2026
- Built a real-time multimodal voice agent tutor that talks students through concepts and problems while drawing diagrams and equations on a live whiteboard or highlighting their uploaded notes.
- Built a custom SVG whiteboard engine with local LaTeX rendering and keyframe animations timed to the agent's speech.
- Built the voice loop: on-device speech detection, ElevenLabs transcription and speech, and interruption-safe turn-taking.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, C++, C, Rust
Frameworks & Libraries: React, Next.js, React Native, FastAPI, Flask, Spring Boot, PyTorch, Anthropic SDK
Tools: PostgreSQL, Supabase, Redis, Docker, Git, CI/CD, Linux, Hugging Face, Claude Code, Cursor

Honors

Ramp Builders Cup (2026): Best Use of Cursor for TicketDodge, which predicts parking ticket odds across New York.
Google x Yale SOM Build with AI Hackathon (2026): 2nd place for Ocura, glasses for the visually impaired.
Prometheus X Catapult Fellow: $5,000 grant. **YC Startup School 2026:** selected from 30,000+ applicants.